

# **Take-home data analysis**

**Business Insight extraction from the WildChat dataset**

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# Research Roadmap

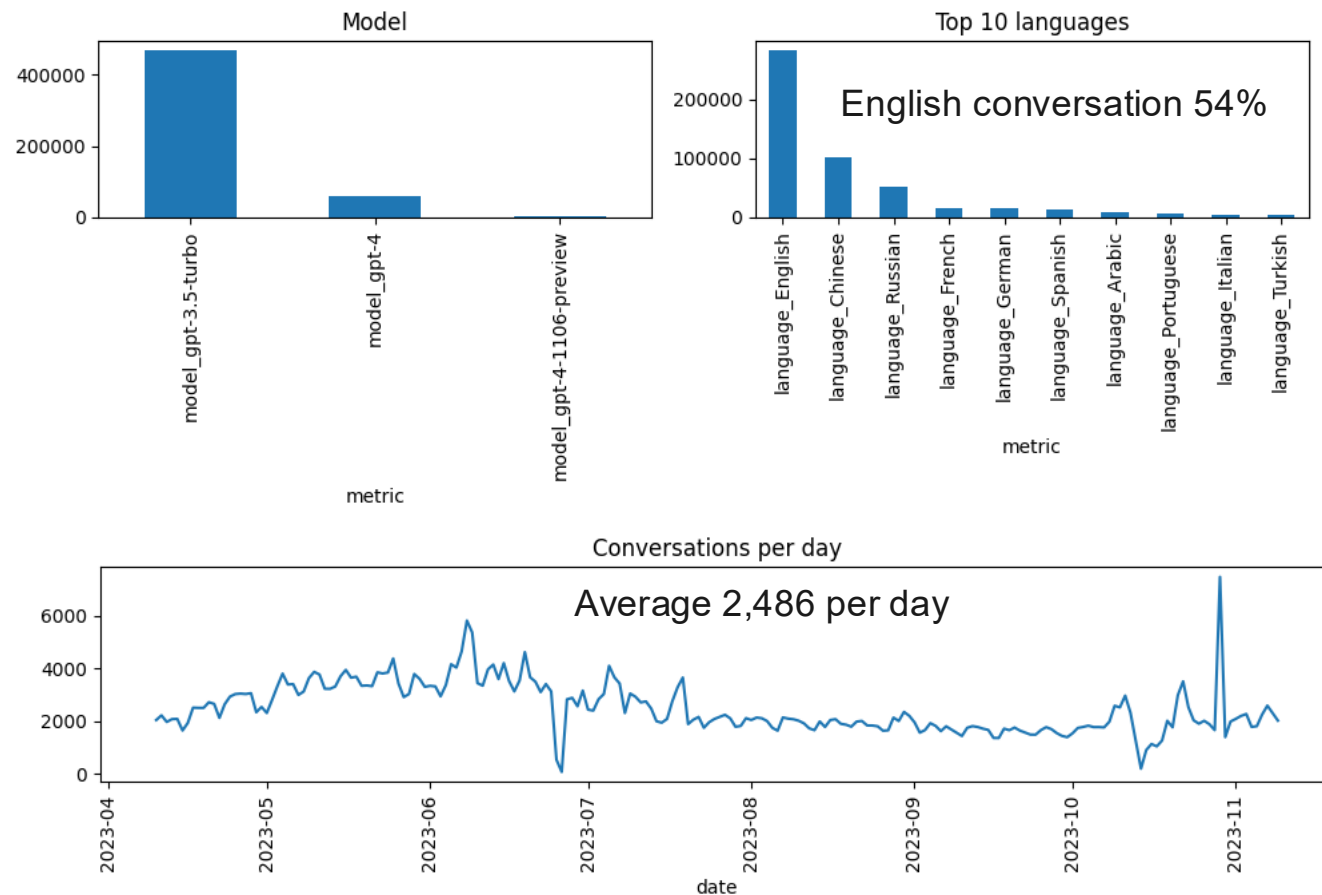
- Dataset overview
- Intent categorization and analysis
- Business vertical and market potential study
- User engagement study across categories and verticals
- Learnings and next steps

# WildChat dataset overview

**WildChat** is a large-scale, real-world conversation dataset released by AllenAI that captures authentic user–AI interactions across languages, intents, and tasks, enabling empirical analysis of usage patterns, behavior, and model performance in the wild.

## Scale & Representativeness

- Average conversations per day: **2,486**
- Estimated total ChatGPT conversations in 2023: **~10M–100M/day**
- Dataset coverage of global usage: **~0.002%–0.025%** (**≈ 0.01% avg**)
- Therefore, any dollar value estimated from this dataset should be scaled by **~10,000×** to approximate total market opportunity.
- If focusing only on English conversations (**54% of data**), scaling increases to **~20,000×**.
- **Next step:** analysis will focus on English conversations for consistency and interpretability.



# **Intent categorization and analysis**

# Understand User Intent Behind AI Conversations

User prompts can be categorized using a standard search-intent framework adapted from SEO and information-retrieval research. This classification helps us understand *why* users interact with AI, whether they’re seeking knowledge, navigation, evaluation, or action, which is critical for product, monetization, and UX analysis.

| Category                 | Definition                                                                                                                                  |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Informational            | The user wants to learn something, get help, or have the AI generate or fix content (explanations, writing, coding, troubleshooting, etc.). |
| Navigational             | The user is trying to find a specific site, app, organization, or resource.                                                                 |
| Commercial investigation | The user is researching or comparing options before making a purchase decision.                                                             |
| Transactional            | The user intends to complete a real-world action such as buying, booking, subscribing, or requesting a service.                             |

**Commercial investigation** and **transactional** queries typically signal stronger purchase intent and therefore represent higher monetization potential for brands than informational conversations. In the following sections, we’ll dive deeper into these high-value intents and quantify their business impact.

# Detailed Intent Distribution of English Conversations

After classifying conversations into high-level intent categories, we further segment the dataset into subcategories to understand how users actually use AI. The percentages shown are calculated from a base of approximately **280K English conversations**, which provides the foundation for this distribution analysis.

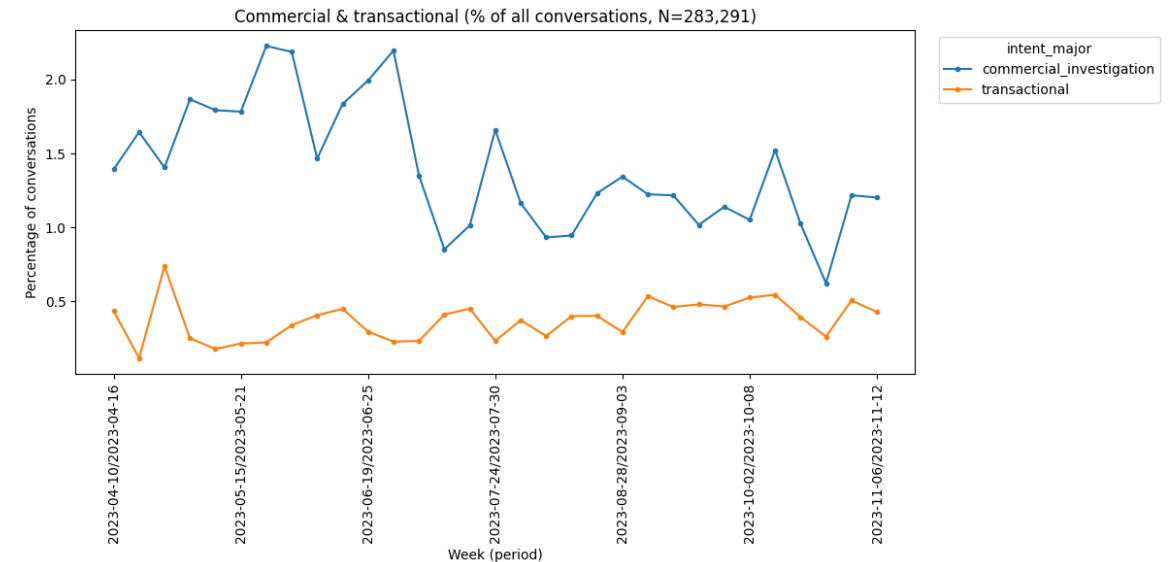
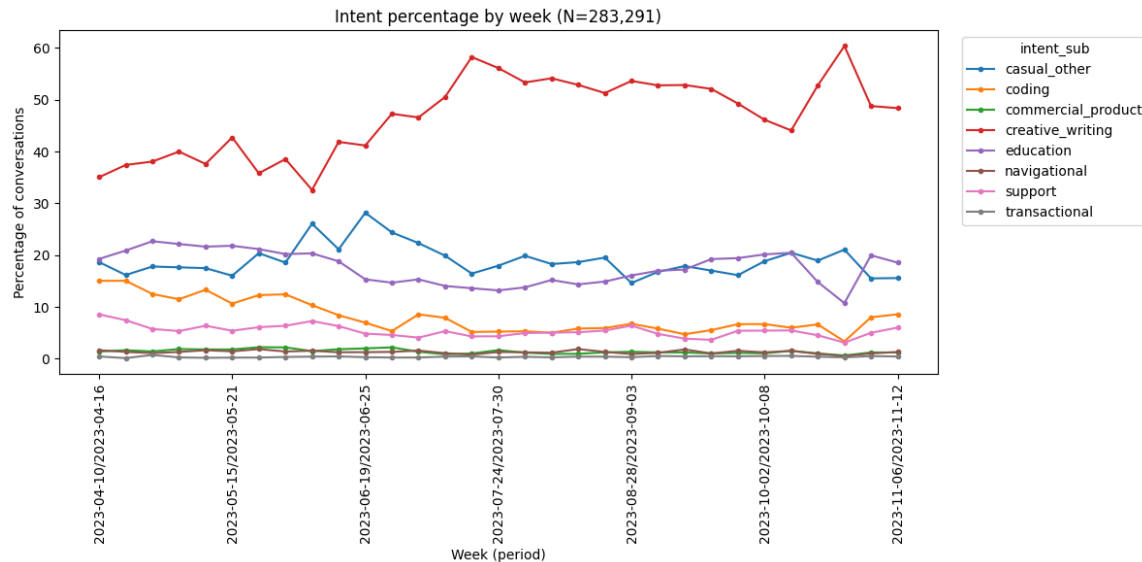
| Category                 | Subcategory             | Conversations | % of Dataset | Definition                                                      |
|--------------------------|-------------------------|---------------|--------------|-----------------------------------------------------------------|
| Informational            | <b>Creative Writing</b> | 130,455       | 46.6%        | Generating stories, scripts, dialogue, or imaginative text.     |
| Informational            | <b>Casual / Other</b>   | 54,774        | 19.6%        | General questions, light chat, or uncategorized requests.       |
| Informational            | <b>Education</b>        | 50,217        | 17.9%        | Learning concepts, explanations, or study help.                 |
| Informational            | <b>Coding</b>           | 23,774        | 8.5%         | Writing, debugging, or understanding code or technical systems. |
| Informational            | <b>Support</b>          | 15,260        | 5.5%         | Troubleshooting problems or fixing issues.                      |
| Commercial Investigation | <b>Product Research</b> | 4,112         | 1.5%         | Comparing or evaluating products before purchase.               |
| Navigational             | <b>Navigational</b>     | 3,693         | 1.3%         | Finding a specific site, app, or resource.                      |
| Transactional            | <b>Transactional</b>    | 1,006         | 0.4%         | Completing an action like buying, booking, or registering.      |

## Key Insights

- **Creative writing is the single largest use case (130K+)**, showing strong demand for generative content.
- **Technical + learning intents** (coding, education, support) together account for a major share, reinforcing AI’s role as a productivity and knowledge tool.
- **Commercial and transactional queries are rare (<2%)**, indicating monetizable intent is a small but high-value segment.

# High-Intent Queries Drive Disproportionate Brand Engagement

Commercial and transactional queries make up only ~2% of conversations but carry disproportionately high business value, as they reflect users closer to conversion and actively evaluating decisions that can drive purchases or sign-ups. In the following analysis, we compare **commercial**, **transactional**, and **informational (education)** intents to quantify these differences.



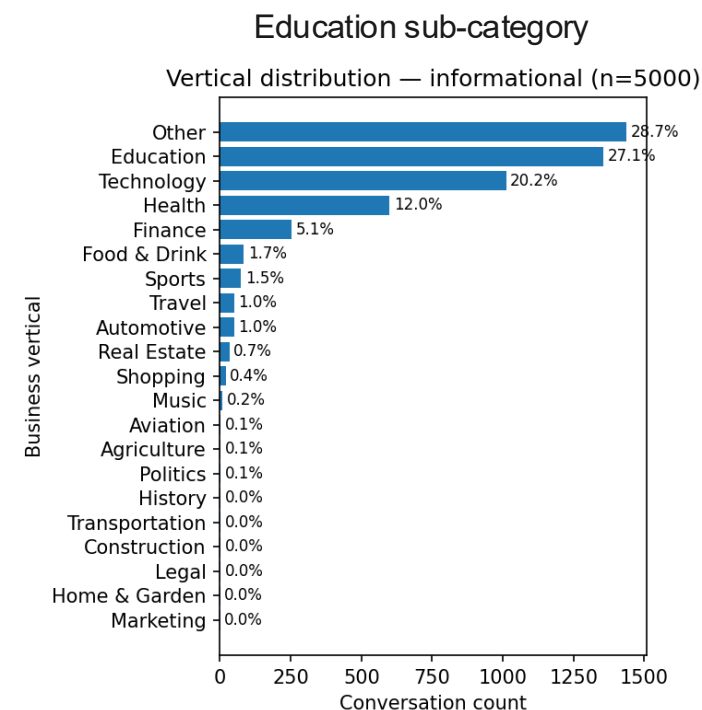
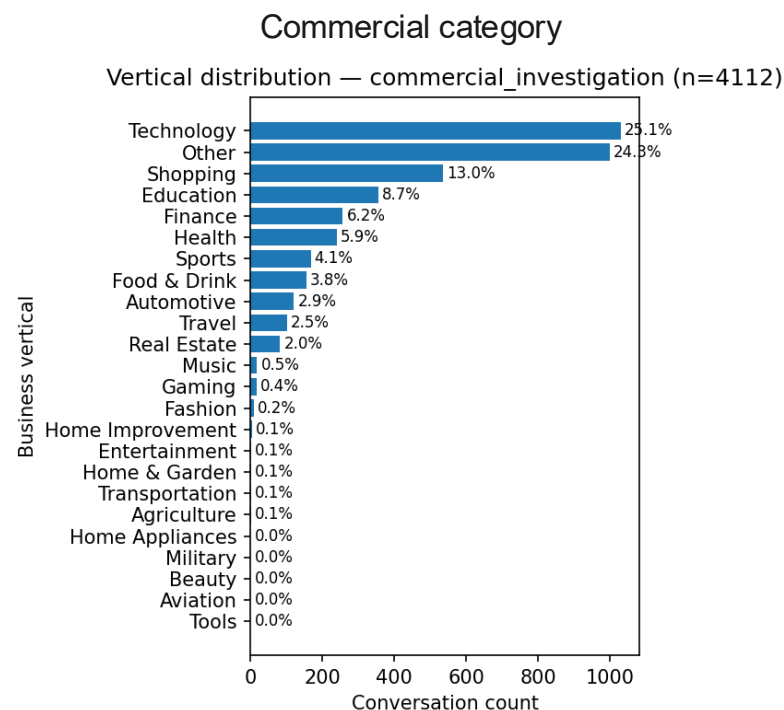
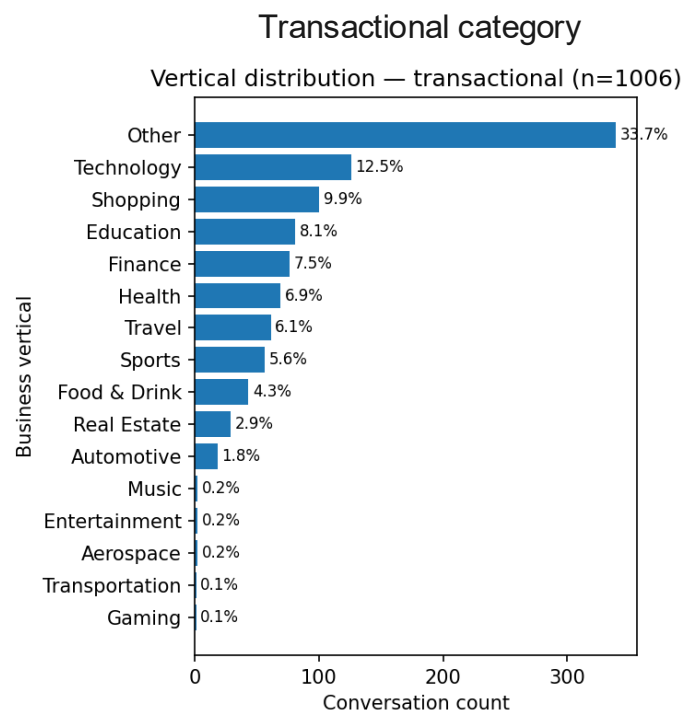
- **Commercial investigation (55.7%)** and **transactional (51.0%)** conversations show the highest **brand mention rates\***, indicating strong brand relevance when users are closer to decision-making.
- By comparison, informational conversations (**education** subset: **22.7%**, computed based on a sampled 5000 conversations) mention brands far less often, suggesting earlier-stage or exploratory user intent.

\***Brand mention rate**: defined as the percentage of conversations where a brand name, entity name or similar is mentioned.

# **Business vertical study**

# Industry Vertical Distribution Across Intent Types

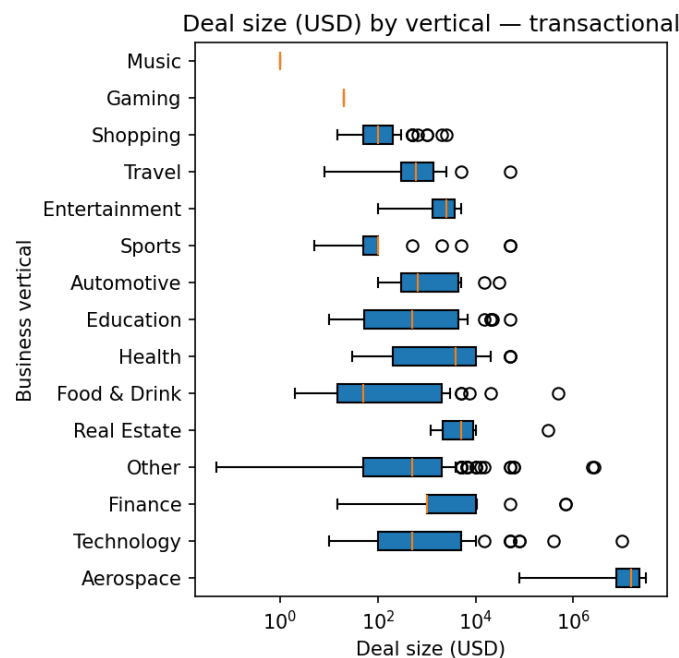
- **Technology consistently ranks among the top verticals across all intent types**, reinforcing it as the most active domain for both research and action-oriented queries.
- **Commercial investigation queries are concentrated in product-driven verticals** such as technology and shopping, aligning with earlier findings that these intents reflect users evaluating purchase decisions.
- **Transactional queries show a broader vertical spread** (finance, health, travel, education), suggesting real-world action intent spans multiple industries rather than being confined to retail.
- **Informational (education-focused) conversations skew toward knowledge-heavy domains** like education, technology, and health, confirming that most AI usage today is still learning- and problem-solving-oriented.
- Compared with volume-dominant informational traffic, **high-intent categories reveal clearer industry signals**, making them more actionable for business and monetization analysis.



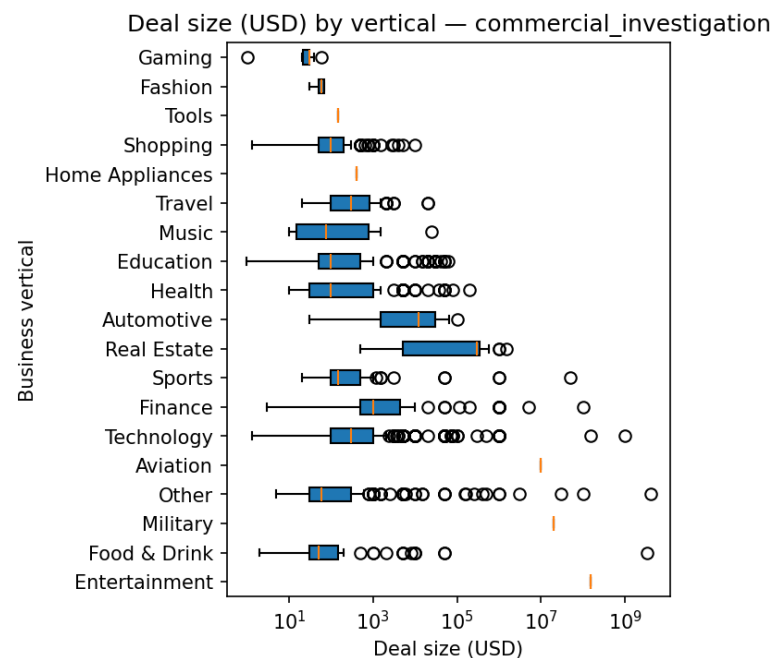
# Estimated Deal Value by Industry Vertical Across Intent Types

- **Higher intent → higher value:** Commercial and transactional queries generally show larger deal sizes than informational, reinforcing that intent quality drives economic potential.
- **Top monetizable verticals:** Technology, Finance, Real Estate, and Automotive consistently exhibit higher typical values and strong outliers. Considering the query volume, **Technology** is clearly the winner.
- **Funnel signal:** Informational queries skew lower-value because they reflect early-stage research, not low-value users.

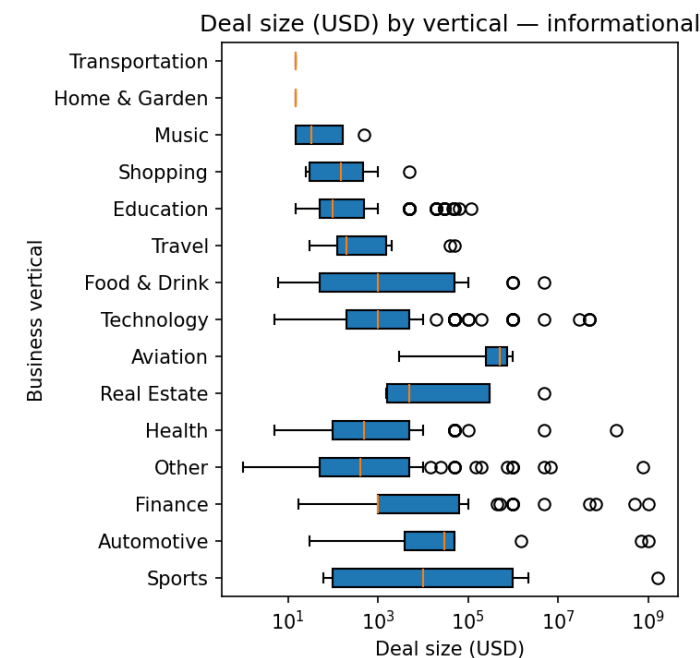
Transactional category



Commercial category



Education sub-category

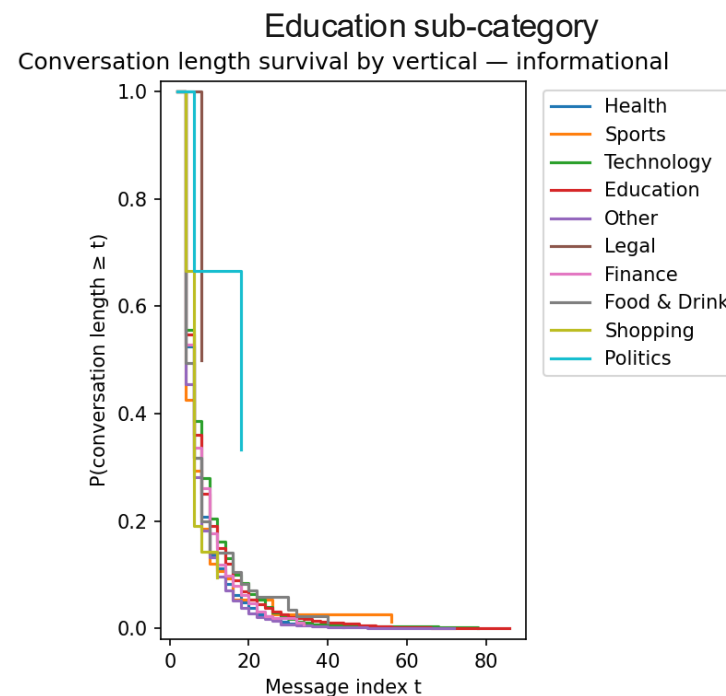
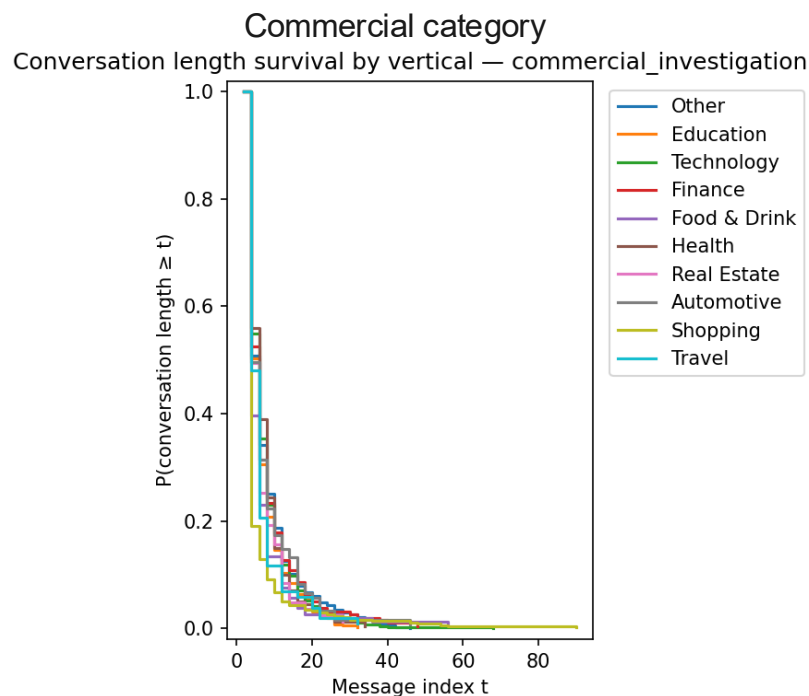
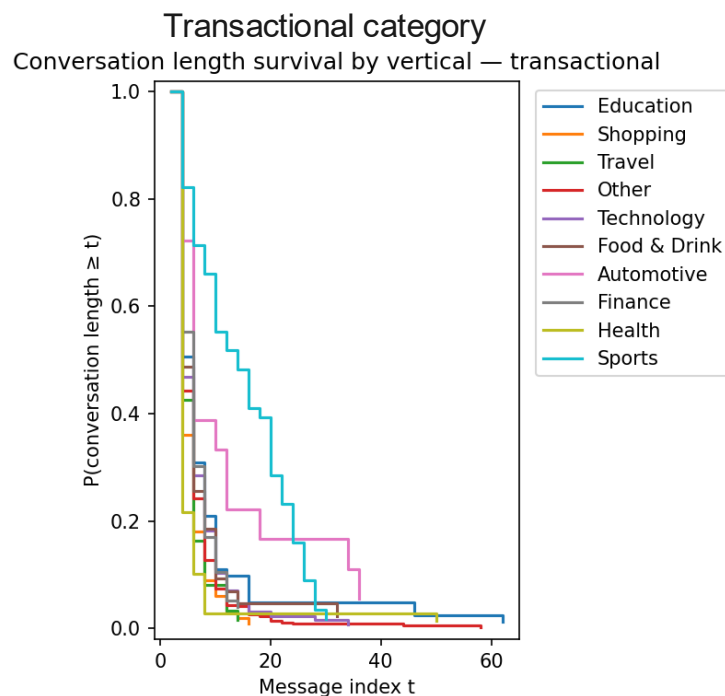


Deal values are **LLM-estimated** based on brands, entities, and products mentioned in conversations. Values are directional rather than exact; some outliers (e.g., aerospace or aviation) may reflect product mentions rather than true purchase intent. Despite noise, the distributions still provide meaningful comparative signals across major consumer and commercial verticals.

# **User engagement study**

# Conversation Survival Curves by Vertical: Engagement Depth Across Intents

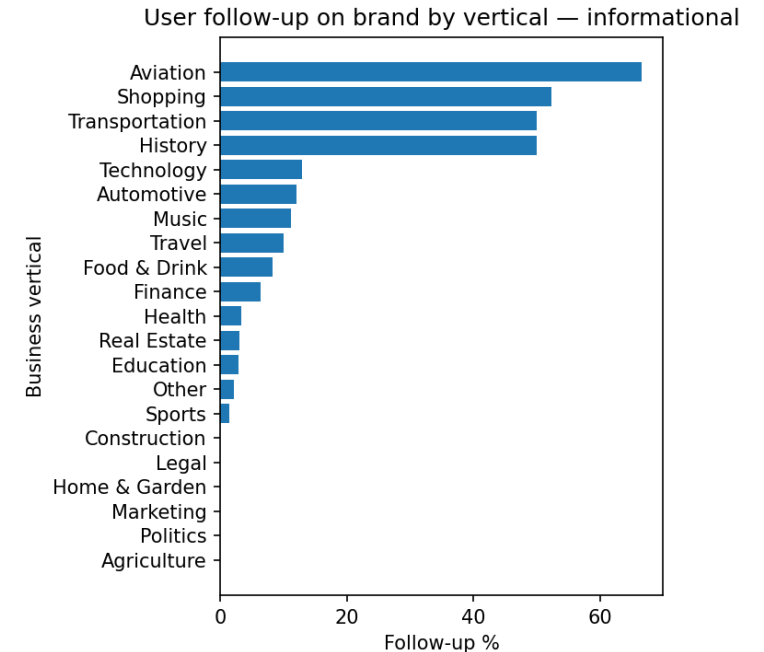
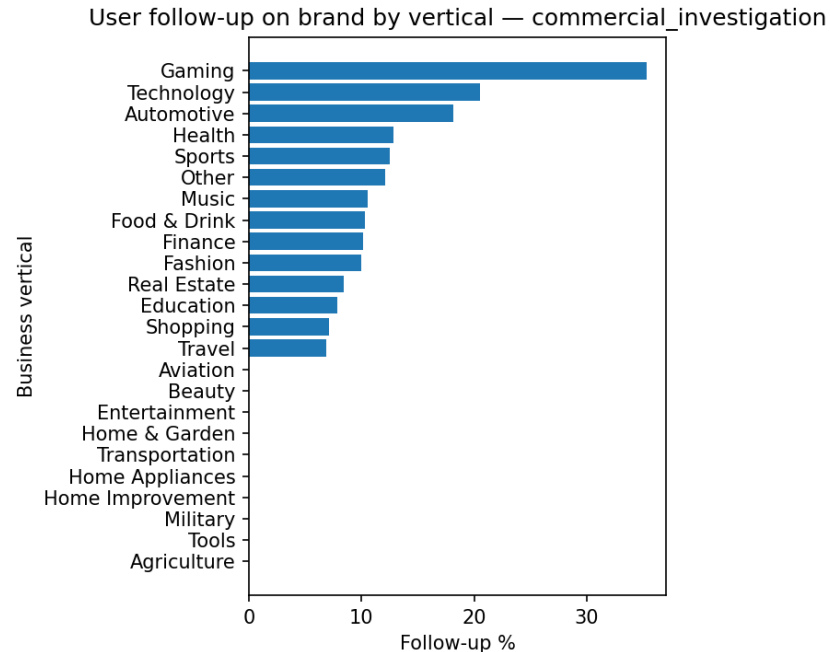
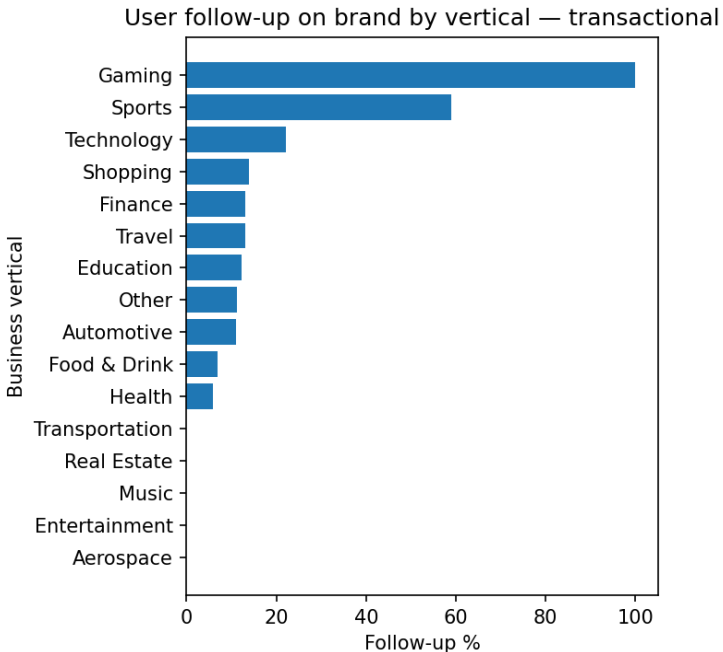
- High-engagement niches may represent **high-value specialization opportunities**, not just large-volume markets.
- **Transactional conversations sustain longer tails** than informational or commercial, indicating higher engagement when users are action-oriented.
- Within transactional: **Sports and Automotive show especially strong retention**, with curves that decay noticeably slower.
- However, these segments have **mid-low query volume**, meaning engagement quality is high despite smaller traffic share.
- Further analysis is needed to understand **why Automotive and Sports users engage longer** (e.g., complexity, decision cost, comparison behavior).



# Brand Follow-Up Behavior by Vertical and Intent

A conversation counts as a *brand follow-up* if the assistant mentions a brand and the user references that same brand in their **next message**, indicating active engagement or interest. This metric captures **brand responsiveness**, not just brand exposure.

- **Technology and Shopping dominate follow-up volume** across commercial and transactional intents, suggesting strong real-world product consideration.
- **Transactional distribution is more balanced across verticals**, indicating brand-driven engagement spans multiple industries when users are action-oriented.
- **Gaming shows strong engagement behavior** (users often continue discussion after brand mention), but prior analysis shows it has **low query volume and smaller deal sizes**, limiting its commercial priority.
- **Education and Health appear consistently across intents**, implying stable cross-funnel interest rather than isolated spikes.



# **Learnings and next steps**

# Closing Takeaways & Strategic Next Steps

## Key Insights

- **High-intent queries (commercial + transactional)** are rare but deliver disproportionate business value and stronger brand engagement signals.
- **Technology stands out as the highest-impact vertical** due to its combination of large query volume, strong brand interaction, and high deal-size potential.
- **Sports and Automotive show unusually deep engagement behavior**, indicating users close to purchase decisions, though current query volume is mid-to-low.
- **Best ROI targets balance three factors:** engagement depth, query volume, and deal value; Technology and Shopping currently score highest across all.

## Learnings & Next Steps

- Manual spot-checking was essential for validating patterns and building intuition about conversational behavior.
- Current labels rely on an **LLM-based pipeline**, fast to deploy but costly and susceptible to noise; deeper validation is needed for precise dollar-value estimation.
- **Technical priority:** develop lightweight embedding-based classifiers for scalable intent and vertical labeling as dataset size grows (LLM labeling will not scale to 100× data).
- **Strategic priority:** map the full user decision journey and value chain to quantify how conversational signals translate into measurable business impact for Profound customers.